

# Sarthak Jain

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Published researcher who makes **AI and signal-processing workloads run fast on resource-limited hardware**: built a pipelined **RISC-V CPU**, ran a quantized neural net on a microcontroller in **under 8 ms**, and aligned brain signals to **under 2 ms**.

## EDUCATION

**Carnegie Mellon University** / *B.S. Electrical & Computer Engineering + Robotics* / **GPA 3.8/4.0** Expected May 2028

- **Coursework**: Logic & Circuit Design (digital systems, HDL), Signals & Systems (Fourier/FFT, filtering, sampling), Imperative Computation (algorithms & data structures), Linear Algebra, Differential Equations, Multivariable Calculus

## RESEARCH EXPERIENCE

**AirLab, Carnegie Mellon** / *Robotics Autonomy Researcher: physics-informed ML* May 2026 – Present

- Turned a drone's own motor and thrust signals into a **proximity sensor**, detecting nearby walls and floors with **zero added sensors** via a **physics-informed PyTorch** time-series model (<3 cm distance, <4° angle error).
- Built an **ANSYS Fluent CFD digital twin** that auto-generated a **120K-sample** training set; closed the sim-to-real gap with domain randomization.

**ForAll Lab, Carnegie Mellon** / *Research Intern: signal processing for a deep-brain BCI* Mar 2026 – Present

- Synchronized a task computer and a neural recorder to **under 2 ms**, so each brain signal lines up with the event that caused it.
- Modeled clock drift with least-squares fitting in **NumPy/SciPy**, cutting timing spread ~30×; de-noised trigger bursts for **100% marker recovery** and automated data validation.

**Kantor Lab, Carnegie Mellon** / *Research Assistant: real-time AI inference on a farm robot* 2025 – Present

- Trained and deployed a real-time **YOLOv8** model that detects and grades fireblight disease in apple trees; team **won the Farm Robotics Challenge (\$25K)**.
- Built the **ROS 2** pipeline turning detections into robotic-arm motion; profiled and fixed vision-to-control latency and sync issues.

**Photovoltaic Efficiency Research** / *Co-Author, with a Harvard PhD researcher* Jan 2023 – Nov 2024

- Simulated solar irradiance across **spectral and angular** dependencies; **co-authored a paper published** in an international renewable-energy journal.

## TECHNICAL PROJECTS

**Edge-AI Health Vest** | *TinyML, INT8 quantization, sensor fusion, Kalman filtering*

- Ran a **pruned, INT8-quantized neural network on a microcontroller** in **under 8 ms** on a **sub-milliwatt** budget, offline.
- Fused **7 biosignal channels**; a denoising autoencoder + adaptive Kalman filter cut motion noise **43%**. Partnered to deploy to **500+ blind athletes**.

**5-Stage Pipelined RISC-V Processor** | *SystemVerilog, computer architecture, Vivado, QuestaSim*

- Designed a CPU from scratch (RV32I) that works on **5 instructions at once**, with hazard-detection and forwarding logic to stay correct without needless stalls.
- Synthesized in **Xilinx Vivado**; verified **cycle-accurately** in QuestaSim with **0 hazard failures**, plus a pipeline visualizer.

**Shadow Fight: Real-Time Vision System** | *C++17, QNX RTOS, OpenCV, dual Raspberry Pi, UDP*

- Wrote low-level **C++17** split across two Raspberry Pis on the **QNX real-time OS** with **under 80 ms** camera-to-screen latency, using a lean ~50-byte **UDP protocol** at 30 fps; cross-compiled with CMake.

**AI Air Hockey Opponent** | *Computer vision, trajectory prediction, stepper motion control*

- Robot tracks the puck, predicts its path (with wall bounces), and moves a 2-axis gantry to block it in **under 50 ms**; profiled the full capture-to-motor loop to remove bottlenecks.

## ENGINEERING & LEADERSHIP

**IEEE x CMU** / *Electrical Lead: animatronic control system* Mar 2025 – Present

- Led a **6-person** team; architected the **RTOS task system in embedded C** coordinating motors, audio, and sensors.

**Fleeca** / *Engineering Intern* Jan 2023 – Jul 2023

- Designed an automated wire cutting/stripping system and custom fixtures, **raising production throughput 22%**.

## TECHNICAL SKILLS

**Languages**: C/C++ (C++17), Python, SystemVerilog/Verilog, Embedded C, Rust, MATLAB

**AI & Inference**: PyTorch, YOLOv8, OpenCV, INT8 quantization, pruning, TinyML, physics-informed ML

**Signal Processing & Math**: Fourier/FFT, filtering, Kalman filtering, state estimation, time-series alignment, NumPy/SciPy, CFD

**Hardware & Architecture**: RISC-V, pipelined CPU design, RTL design/verification (Vivado, QuestaSim), microcontrollers, PCB (Altium)

**Systems & Tools**: Linux, QNX RTOS, ROS 2, multithreading, POSIX sockets, latency profiling, CMake, cross-compilation, Git